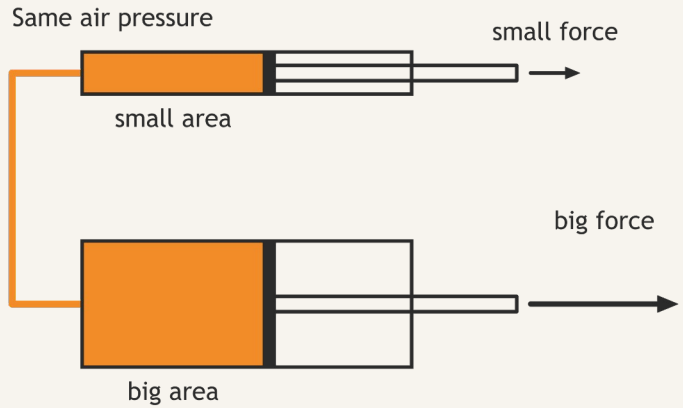
Daily Review

1	The rod moves in. The exam word?	Instroke
2	Double-acting: what pushes the rod back?	Air
3	Arrow to the cylinder: the rod goes out…	Slowly
4	Our safe gauge reading?	1 to 2 bar
5	Area of a 10 mm by 20 mm rectangle?	200 mm^2

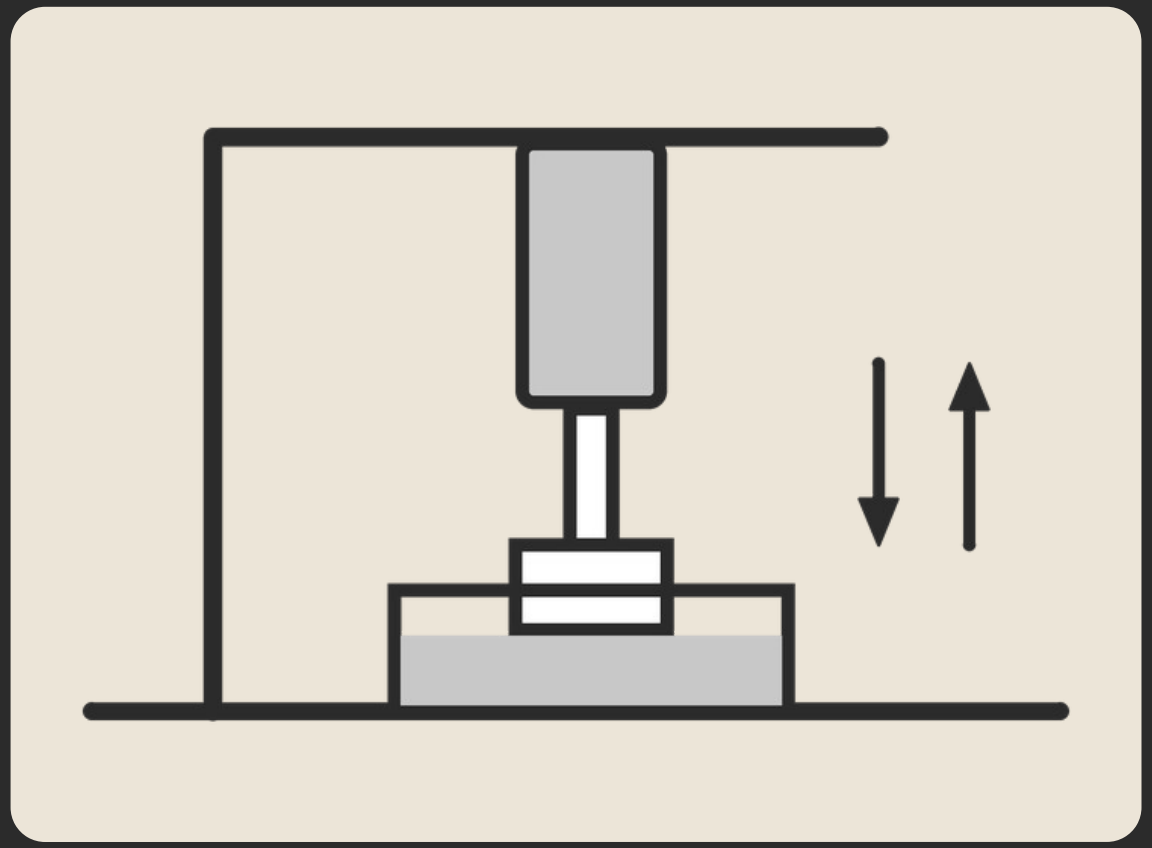
Will It Lift?

How heavy a basket can the air lift?



S3 Engineering Science

Will the air lift the 50 N basket?



Air: 0.2 N mm^{-2} on 300 mm^{2}

By the end of this lesson…



1. I can use P = F/A to find a pressure or a force
2. I can find a piston area from a pressure and a force
3. I can describe how the paint dip works, stage by stage
4. Level Up: I can find an area from a diameter

Why a Bigger Cylinder Pushes Harder

Same pressure

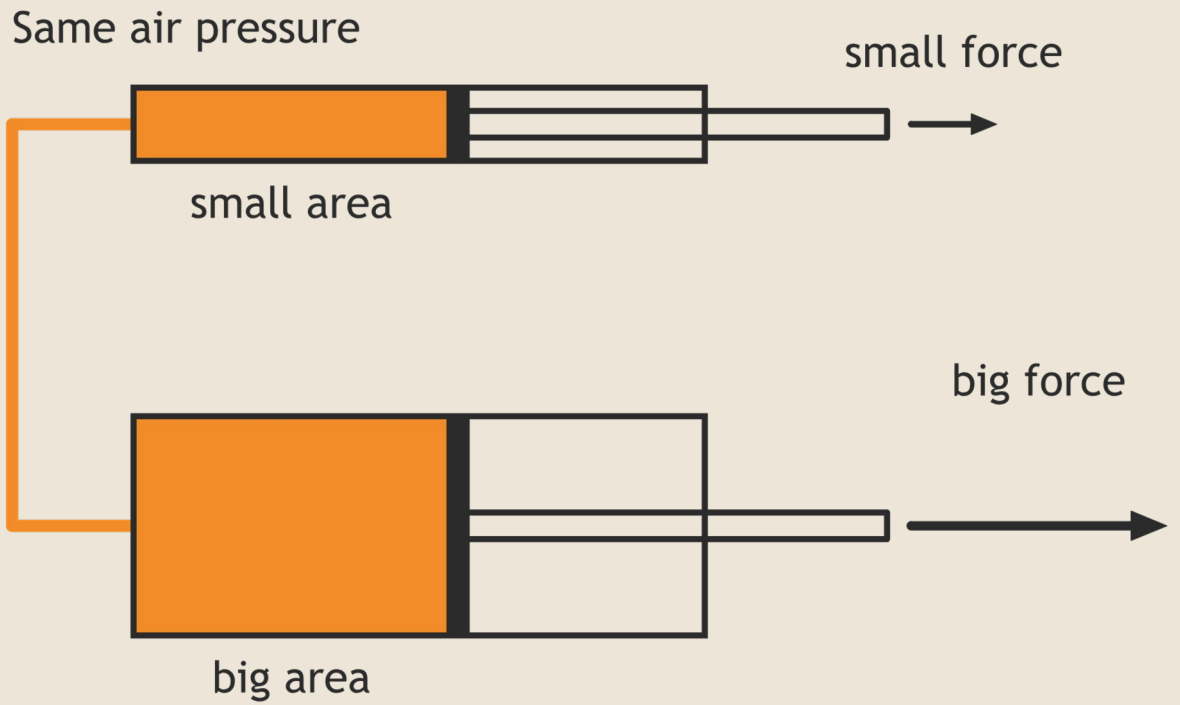
the same push on every mm^2 of piston

Bigger area

more mm^2 for the air to push on

So

a bigger force



Pressure, Force and Area

$$P = F/A$$

P = pressure, in N mm^{-2}

F = force, in N

A = area, in mm^2

Our gauge: 1 bar = 0.1 N mm^{-2}



Worked Example 1: Find the Pressure (2 marks)

A piston has an area of 250 mm^2. It outstrokes with a force of 40 N. Calculate the air pressure supplied.

$F = 40\ N$

$A = 250\ mm^2$

$P = ?$

$P = F/A$

$P = 40/250$

$P = 0.16\ N\ mm^{-2}$

We Do: Find the Pressure (2 marks)

A piston has an area of 300 mm^2. It outstrokes with a force of 54 N. Calculate the air pressure supplied.

$F = 54\ N$

$A = 300\ mm^2$

$P = ?$

$P = F/A$

$P = 54/300$

$P = 0.18\ N\ mm^{-2}$

Your Step: Find the Force (3 marks)

The air pressure is 0.30 N mm^{-2} and the piston area is 200 mm^2. Calculate the outstroking force.

$P = 0.30\ \text{N mm}^{-2}$

$A = 200\ \text{mm}^2$

$F = ?$

$P = F/A$

$0.30 = F/200$

$F = 0.30 \times 200$

$F = 60\ \text{N}$

Worked Example 2: Find the Area (3 marks)

Cylinder A is supplied with an air pressure of 0.40 N mm^{-2} and outstrokes with a force of 180 N. Calculate the piston area.

$P = 0.40$ N mm^{-2}

$F = 180$ N

$A = ?$

$P = F/A$

$0.40 = 180/A$

$A = 180/0.40$

$A = 450$ mm^2

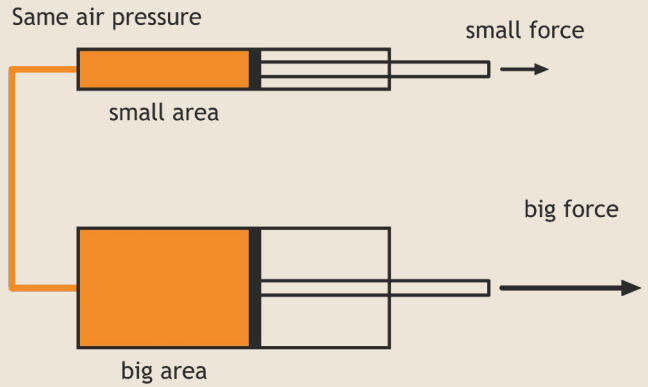
Your Turn

Show Me board

1 F = 90 N, A = 300 mm^2. Find P.

2 P = 0.25 N mm^{-2}, A = 320 mm^2. Find F.

3 P = 0.50 N mm^{-2}, F = 210 N. Find A.



5 min

Answers: Your Turn

1 1 $P = 90 \div 300$

2 2 $F = 0.25 \times 320$

3 3 $A = 210 \div 0.50$

$0.30\ N\ mm^{-2}$

80 N

$420\ mm^2$

Back to the Job: Will It Lift?

1. $P = 0.2$ N mm^{-2}, $A = 300$ mm^2. The force?

   60 N

2. The basket weighs 50 N. Will it lift?

   Yes: 60 N is more than 50 N

3. Gauge down to 1 bar. Will it lift now?

   No: only 30 N

Exam Practice (2 marks)

The piston in the cylinder has a cross sectional area of 420 mm^2 and instrokes with a force of 65 N. Calculate the air pressure supplied to instroke the piston.

$$P = F/A$$

$$P = 65 \div 420 \quad (1)$$

$$P = 0.15 \text{ N } mm^{-2} \quad (1)$$

Describe the Operation

One line per stage

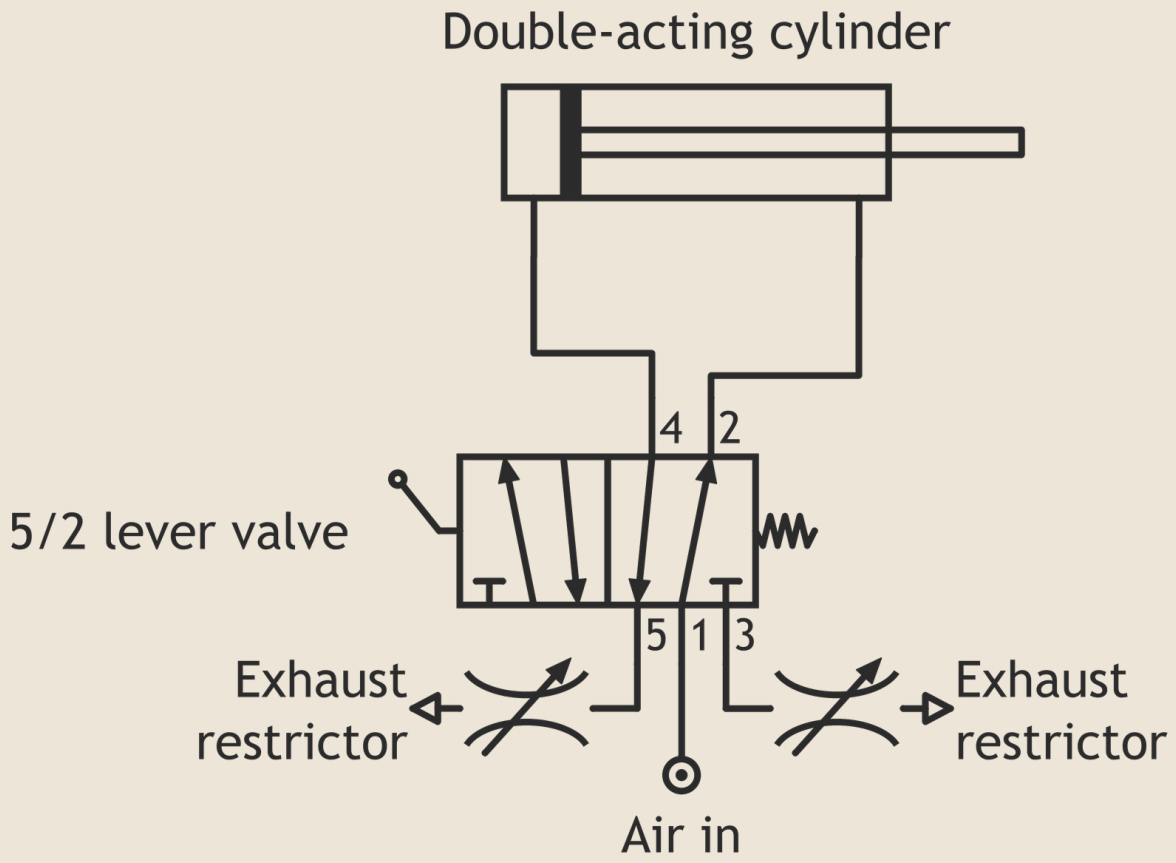
one mark each

Say what changes

the lever is actuated; the valve changes state

Say what the piston does

it outstrokes or instrokes

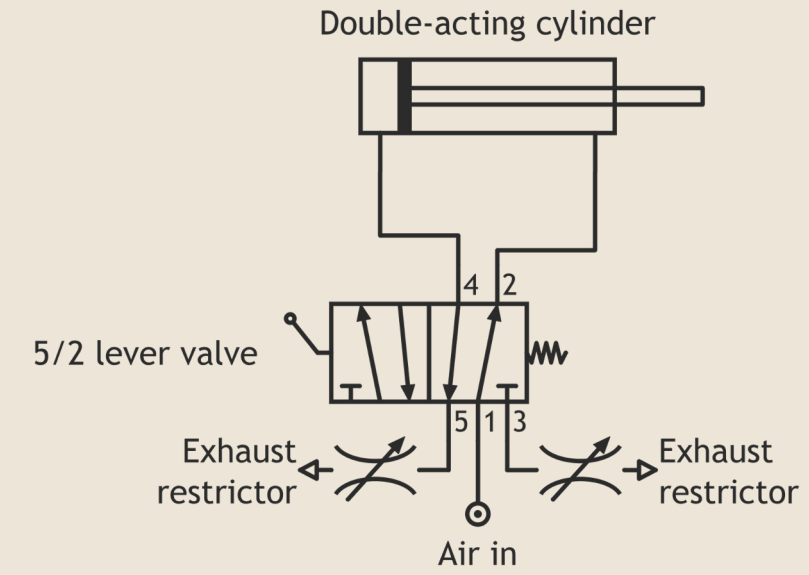


Show Me: Describe the Dip (2 marks)

When the lever is actuated …

Stage 1 (1): the 5/2 valve changes state and the piston outstrokes. The basket dips.

Stage 2 (1): when the lever is released, the spring resets the 5/2 valve and the piston instrokes. The basket lifts.



Level Up: Area from a Diameter

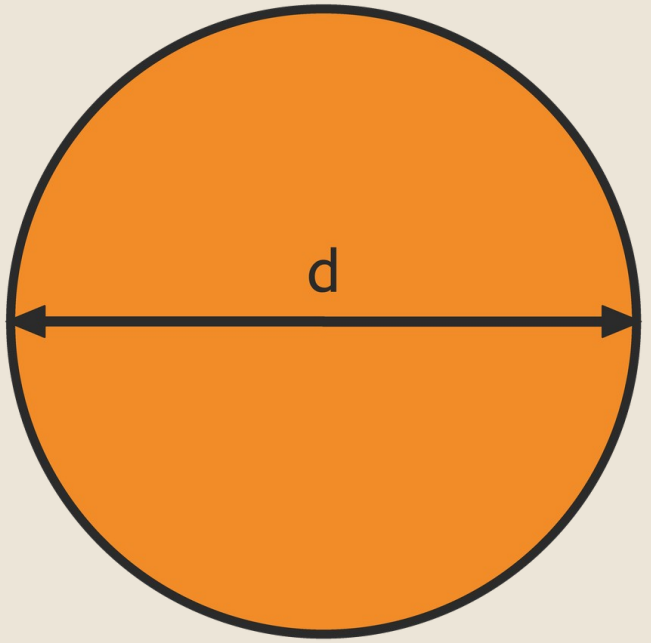
Level Up

$$A = (\pi d^2)/4$$

A = area, in mm^2

d = diameter, in mm

π: the π button



d = diameter

Worked Example 3: The Force (4 marks)

Level Up

The air pressure supplied is 0.20 N mm^{-2}. Calculate the outstroking force of a 30 mm diameter piston.

d = 30 mm

P = 0.20 N mm^{-2}

F = ?

$A = (\pi \times 30^2)/4 = 706.9$

$0.20 = F/706.9$

$F = 0.20 \times 706.9$

$F = 140$ N

Level Up: Exam Practice (3 marks)

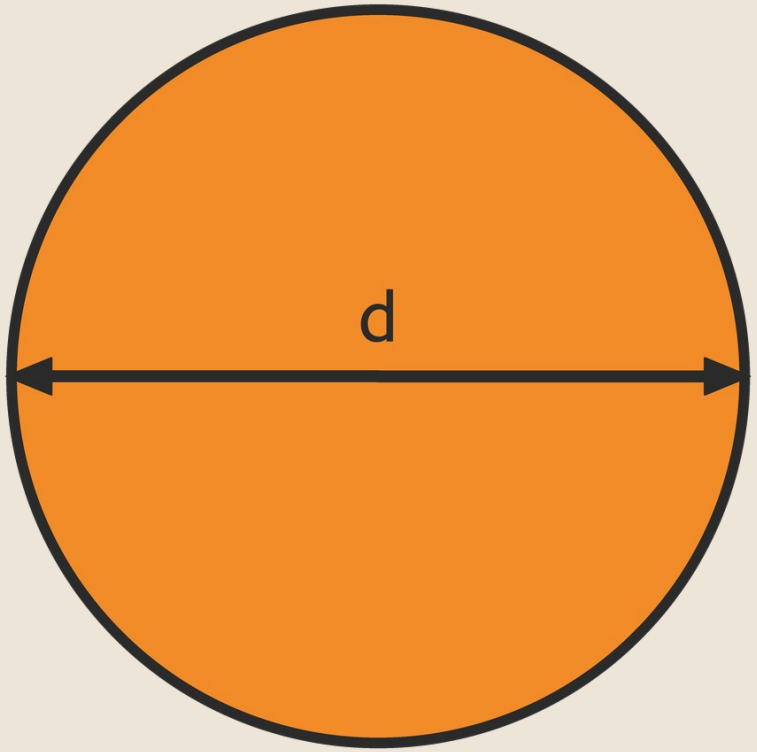
Level Up

Cylinder B requires an outstroking force of 18 N. Calculate the air pressure required if the diameter of the piston is 12 mm.

$A = \pi \times 12^2 \div 4 = 113\ (mm^2)$ (1)

$P = 18 \div 113$ (1)

$P = 0.16\ N\ mm^{-2}$ (1)



d = diameter

Level Up: Out Strong, Back Weaker

Level Up

Cause (1)

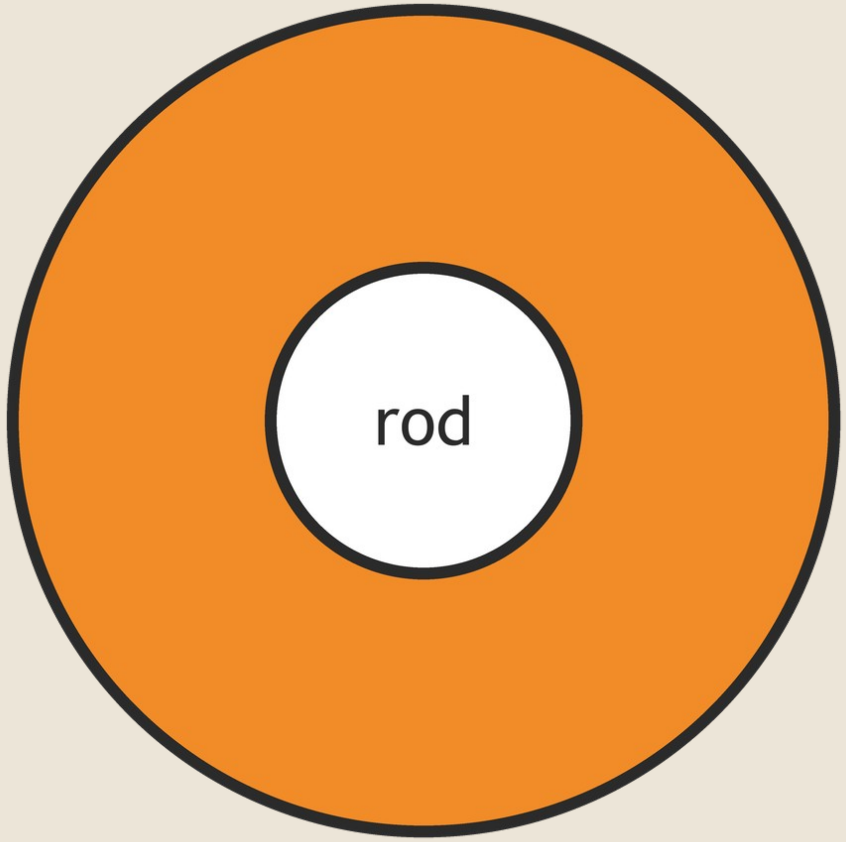
the rod makes the instroke area smaller

Effect (1)

so the instroke force is smaller

Watch Out

say area, not size



Air pushes on the ring

Worked Example 4: The Ring (3 marks)

Level Up

A piston is 40 mm in diameter. Its rod is 10 mm in diameter.

Calculate the effective area of the piston when it instrokes.

$D = 40\ \text{mm}$

$d = 10\ \text{mm}$

$A = ?$

$A = (\pi \times 40^2)/4 - (\pi \times 10^2)/4$

$A = 1256.6 - 78.5$

$A = 1200\ \text{mm}^2$

Plenary: Will It Lift?

1	P = F/A. What is A?	Area, in mm^2
2	F = 60 N, A = 300 mm^2. P?	0.20 N mm^{-2}
3	Same pressure, bigger area: the force is…	Bigger
4	Describe: one mark for each…	Stage: valve and piston
5	Level Up: why is the instroke weaker?	The rod: smaller area

Look back — by the end of this lesson…



1. I can use P = F/A to find a pressure or a force
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4. Level Up: I can find an area from a diameter

LOOKING AHEAD

Lesson 6: Maintenance Night

Rebuild the dip, hide a fault, find one.

Credits (1 of 2)

Drawings: Sources: job card, cylinder and piston drawings by R Stewart.

Worked example layout: © Qualifications Scotland (SQA). Worked examples set out as the N5 Engineering Science marking instructions set out 2021 Question 12(c), 2022 Question 15(c) and 2025 Question 8(d); numbers are new.

Exam Practice: 2026 Q14(b): © Qualifications Scotland (SQA). N5 Engineering Science 2026, Question 14(b).

Describe the operation: © Qualifications Scotland (SQA). Describe-the-operation layout from the N5 Engineering Science marking instructions (2021 Q12(a), 2023 Q9(a), 2024 Q15(a), 2025 Q8(b)).

Lesson 4 circuit: The Lesson 4 circuit adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 4 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Exam Practice: 2024 Q15(b): © Qualifications Scotland (SQA). N5 Engineering Science 2024, Question 15(b) and its marking instructions.

Credits (2 of 2)

Out Strong, Back Weaker: 2022 Q15(d): © Qualifications Scotland (SQA). N5 Engineering Science 2022, Question 15(d) and its marking instructions.

R Stewart - October 2026